**Problem Set (5.2)**

5.2.1 Evaluate the following determinants:

(a) $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$, (b) $\begin{vmatrix} 5 & -2 \\ 3 & 1 \end{vmatrix}$, (c) $\begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix}$,

(d) $\begin{vmatrix} 2 & 0 \\ -5 & 1 \end{vmatrix}$, (e) $\begin{vmatrix} 2 & -10 \\ 3 & -15 \end{vmatrix}$, (f) $\begin{vmatrix} 2 & -2 & -1 \\ 3 & 1 & -1 \\ 4 & 3 & 5 \end{vmatrix}$,

(g) $\begin{vmatrix} 2 & 5 & 0 \\ 0 & 3 & 4 \\ -5 & 3 & 6 \end{vmatrix}$, (h) $\begin{vmatrix} 3 & 2 & 1 \\ 1 & -2 & 4 \\ 4 & 2 & 3 \end{vmatrix}$, (i) $\begin{vmatrix} 4 & -3 & 2 \\ 5 & 9 & -7 \\ 4 & -1 & 7 \end{vmatrix}$.

5.2.2 Solve, using determinants:

(a) $x+2y=-4$
$5x+3y=1$

(b) $ax-2by=c$
$2ax-3by=4i$

(c) $3x+2z=8-5y$
$3y+2x=z+1$
$3z-1=x-2y$

(d) $2x-5y+2z=7$
$x+2y-4z=3$
$3x-4y-6z=6$

(e) $2z+3=y+3x$
$x-3z=2y+1$
$3y+z=2-2x$

(f) $2x-3y+2z=6$
$x+8y+3z=-31$
$3x=2y+z=-5$.

5.2.3 Solve, using adjoint matrix:

$x+2y+2z=4$
$3x-y+4z=25$
$3x+2y-z=-4$.

**Problem Set (5.3)**

5.3.1 Find the inverse of the matrix:

(i) $A=\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, (ii) $B=\begin{pmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 0 & 5 & -1 \end{pmatrix}$,

(iii) $C=\begin{pmatrix} 2 & -3 & 4 \\ 1 & 2 & -3 \\ -1 & -2 & 5 \end{pmatrix}$, (iv) $D=\begin{pmatrix} 4 & -1 & -2 \\ 0 & 2 & -3 \\ 5 & 2 & 1 \end{pmatrix}$.